



HORIZON

**SOVEREIGN AUTONOMOUS ENTERPRISE INFRASTRUCTURE
RECOVERY PLATFORM**

THINK PAGERDUTY PLUS AN AUTOPILOT, BUILT IN INDIA, WITH PROOF OF EVERY ACTION

PRESENTED BY FALCONIZERS - 1048

Overview

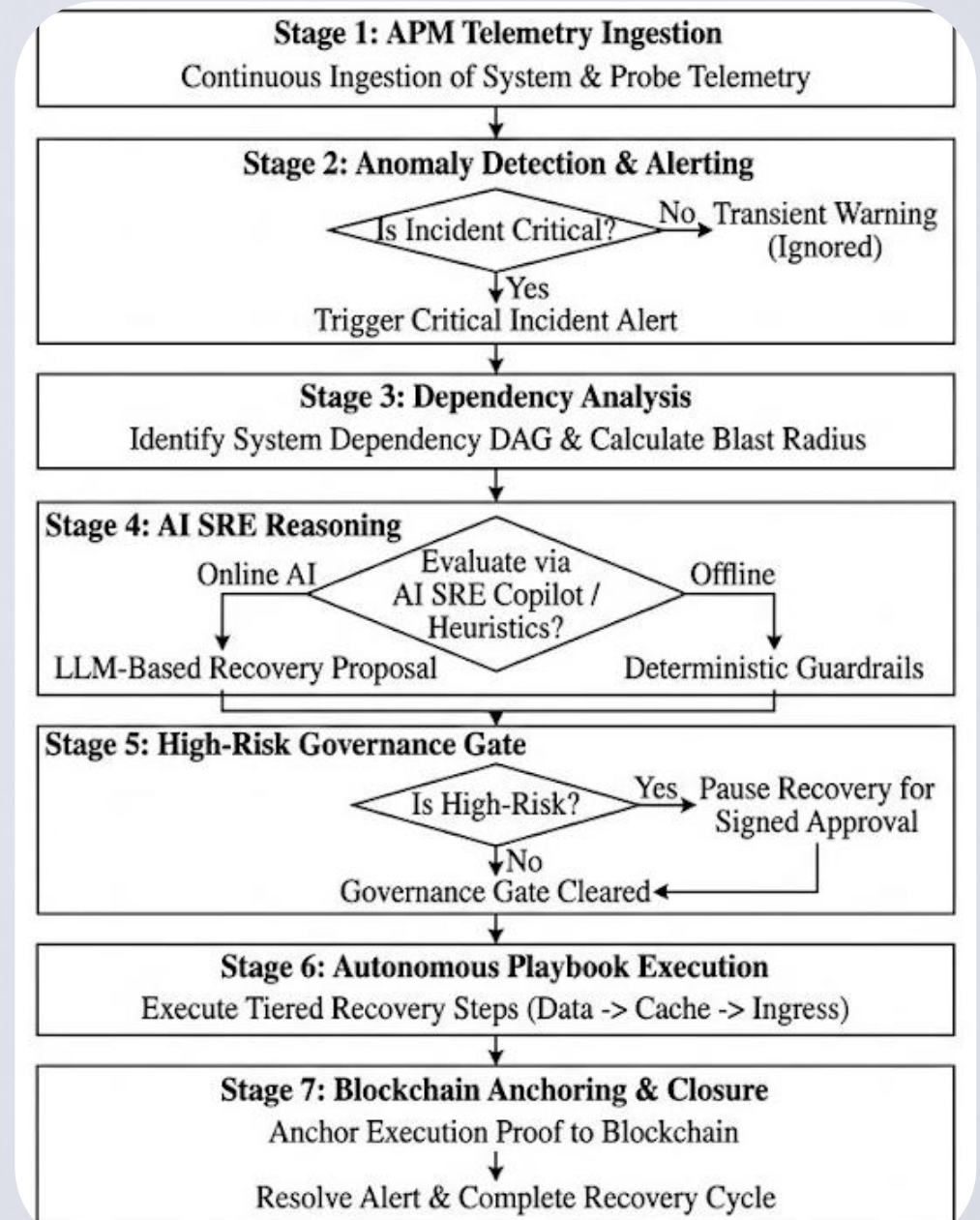
- ❖ **Problem:** Distributed systems fail in chains, while manual recovery and incorrect restart sequences extend costly downtime.
- ❖ **Idea:** Detect failures, map dependencies, and autonomously restore services in the mathematically correct order.
- ❖ **USP:** Combines graph-based recovery, human approval for high-risk actions, and blockchain-backed proof of every action.
- ❖ **Innovation:** Deterministic recovery algorithms, multilingual AI assistance, and tamper-evident audit trails in one platform.

HORIZON



Technical Approach

- Maintains live service DAGs in `ClusterStateManager` and confirms outages using a 3-consecutive-miss probe to eliminate flapping.
- Traverses reverse dependency edges via `computeBlastRadius` in $O(V + E)$ time to isolate the exact downstream cascade before acting.
- Derives bottom-up recovery order via `getTopologicalOrder` (Databases → Caches → APIs) to eliminate thundering-herd crash loops.
- Runs low-risk heals automatically while pausing hazardous mutations for signed wallet approvals via `mstBlockchain`.
- Verifies node health in `Pipeline Deployer` before unlocking dependants and anchors SHA-256 Merkle proofs to `HorizonAuditVault`.



Working Deployed Prototype

Deployed Link: horizon-aiops.vercel.app

Github Repo: [Horizon Repository](https://github.com/bridgekey/horizon-aiops)

Horizon / Specifications / Model Context Protocol (RFC-MCP-2024-11-05)

Living Standard • Production

MST Chain: 91562837

MON WEB DOCS SPECIFICATION • API REFERENCE

Model Context Protocol (MCP) Reference

The Horizon Model Context Protocol (MCP) Server provides standard, bidirectional JSON-RPC 2.0 tool execution for LLM agents. It exposes live dependency DAG inspection, chaos simulation, topological Kahn recovery orchestration, and cryptographic EIP-712 human commander signing gates.

IMPORTANT: Cryptographic EIP-712 Consensus Barrier

High-risk Tier 0 infrastructure recovery operations (database failovers, stateful volume resets, and DNS traffic switchovers) unconditionally require an authorized Commander signature via BridgeKey Wallet on MST Testnet. Autonomous agents cannot execute cutovers without this cryptographic proof.

MCP Server & Tools Reference

Slash (/) Commands

CLI Reference

REST & Edge API

IN THIS GUIDE:

1. Quick Start

2. Client Setup

3. Tool Catalogue

4. JSON-RPC 2.0 Wire

5. Error Codes

6. Compatibility Matrix

SPECIFICATION DETAILS

RFC-MCP-2024

Transports: stdio | SSE | HTTP

Wire Protocol: JSON-RPC 2.0

MST Testnet: Chain 91562837

Auth & Gate: EIP-712 Multi-Sig

Contract: 0x3E0ad23...6887

Launch Agent Architect

Resilience Command Center

Chaos Sentinel

Observability & PagerDuty

AI Architect

MCP & Docs

LIVE SRE ENGINE

Refresh

Onchainstate

Real-time DAG telemetry - AI triage - MST on-chain audit - Last polled 5:34:58 AM

Sarvam AI Autonomous SRE Cogitator

Continuous natural language prompts orchestrating autonomous agents across infrastructure.

ACTIVE MULTI-AGENT

Type SRE command (e.g., 'simulate postgres outage', 'diagnose cluster')...

Execute

QUICK PROMPTS: Simulate outage on PostgreSQL Primary

Diagnose cluster health and blast radius

Execute bottom-up recovery playbook

Check MST testnet signer status

Execute

MEAN TIME TO RECOVERY

18.9s

Optimal SLA

3 incidents auto...

CLUSTER AVAILABILITY

100.0%

All Operational

99.95% Month...

MONITORED SERVICES

K8S & DB

6

PostgreSQL • Redis • Gateways

ACTIVE OUTAGES

0

No Outages

Continuous health...

Infrastructure Services Matrix

6 NODES ACTIVE

All (4)

Data Tier

App Services

Incident Triage Stream

Cryptographically hashed audit ledger

LIVE FEED

Web3 Token-Gated Subscription Plans

Zentrif Pass (ZXPASS) • MST Blockchain Testnet

Mint non-fungible subscription tokens on the MST Blockchain (Chain ID 91562837). Smart contracts autonomously verify authorization tiers without centralized payment gateways.

Active On-Chain Subscription: Guardian Pass

On-chain NFT Pass verified on MST Blockchain Testnet.

Add NFT #3 to BridgeKey

View #3 on MSTScan

Quick Prompts: Live Video Streaming (Ceyllob • FFmpeg)

Live Ticket Booking (CockroachDB • Redis)

Blogging Website (Postgres • Yarnix)

4-Commerce (MySQL • Redis • Stripe)

GenAI RAG (qvector • vLLM)

FinTech Ledger (Kafka • Hentle)

Microservices (K8S)

OTT / Streaming (K8S)

K8S Orchestration (K8S)

Content CMS (K8S)

AI / LLM RAG (K8S)

FinTech Trad... (K8S)

Cloud Baseline (K8S)

Minimal 3-Tie... (K8S)

Custom Blank... (K8S)

pipeline.yaml

apiVersion: horizon.recovery.io/v1alpha1

kind: AutonomousRecoveryPipeline

metadata:

name: live-ticket-booking-flash-sale

environment: production

description: High-concurrency live concert ticket booking mesh with distributed seat locks & queue workers

spec:

governanceData:

required: true

standard: EIP-712

chainId: 91562837

contract: "0x3E0ad238dFcd83C3749b9feda9439646887"

nodes:

- id: cockroach-ledger

name: CockroachDB Seating Inventory Ledger

Valid DAG: 6 Nodes • 5 Topological Recovery Tiers • Acyclic (0 Cycles)

Algorithm: Kahn D (V+E)

Ledger Integrity

100%

Tamper-Evident History

Blockchain Network

MST Testnet

Chain ID: 91562837

Hash Mechanism

SHA-256

Merkle Root Chained

Immutable Audit Trail & On-Chain Hashes

Cryptographically anchored state transitions and human commander approvals.

Search actions or actors...

Timestamp

Actor

Action Executed

Details

10/10/2026, 4:13:52 AM

SYSTEM

Autonomous Health Probe Sync

All 7 nodes reporting 200 OK across

10/10/2026, 4:13:52 AM

HUMAN (Commander)

MST Testnet Validator Ping

Verified MST Testnet RPC connect

ALL 6 NODES VERIFIED WITH CRYPTOGRAPHIC CHECKSUMS (100% GREEN)

Verified Green: 6 / 6

#1 CockroachDB Se...

#2 Redis Distrib...

#3 RabbitMQ High...

#4 Seat Allocated...

#5 Stripe Checkou...

#6 AWS CloudFront...

Feasibility and Viability



- O(V + E) Scalability: Computes 1,000+ node recovery in < 3 ms via DependencyGraph without heavy compute.
- Deterministic Core: Executes recovery via pure TypeScript in PipelineDeployer, isolated from LLM hallucinations.
- Native Connectors: Plugs directly into Docker via dockerBridge, Kubernetes, and standard MCP IDE tools. Cycle & Flap Resilient: Built-in DFS cycle detection stops deadlocks, while sliding-window probes prevent flapping.
- Low-Cost Web3: Hardware-backed EIP-712 wallet gates in mstBlockchain anchor on-chain for < ₹0.10 gas.

- ❖ High ROI: Slashes \$50k/min downtime losses by cutting MTTR from 45 mins to < 2 minutes.
- ❖ Regulatory Driver: Automates India's mandatory 6-hour CERT-In reporting via on-chain HorizonAuditVault.
- ❖ 80%+ Gross Margins: High-margin SaaS model with < ₹12 marginal compute and gas cost per recovery.
- ❖ PLG Flywheel: Developers adopt @horizon/mcp-server inside Cursor and Claude for organic bottom-up sales.
- ❖ Sovereign Advantage: INR pricing removes USD forex volatility, with 100% Indian data residency for BFSI.

Marketing strategies

- ❖ Datadog / Dynatrace: Only alerts on issues; Horizon actuates automated recovery.
- ❖ PagerDuty / Rootly: Pages on-call humans; Horizon resolves incidents autonomously.
- ❖ StackStorm / Rundeck: Runs static scripts; Horizon derives dynamic DAG plans.
- ❖ Generic LLM Agents: Hallucinates commands; Horizon enforces deterministic math.
- ❖ Patent US12556441B1: Static mapping; Horizon re-plans on health probe failures.

Business Model Canvas

- ❖ Core Value: Slashes MTTR by 95%, eliminates out-of-order crash loops, and guarantees tamper-proof audit trails.
- ❖ Target Market: Indian cloud scale-ups, regulated FinTech/Neobanks (CERT-In 6h compliance), and HealthTech.
- ❖ Revenue Model: Predictable INR SaaS subscriptions (₹0 to ₹24,999/mo) + high-ticket Enterprise annual contracts.
- ❖ Unit Economics: 82% gross margins with < ₹12 marginal serverless and blockchain cost per recovered incident.

Thank you

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